



KOLEJ MATRIKULASI MELAKA
UNIT MATEMATIK
SEMESTER I, SESI 2022/2023
MATHEMATICS 1 SM015
KARISMATIK E



ANSWER ALL QUESTIONS

SECTION A [25 marks]

BAHAGIAN A[25 markah]

1. Given that $f(x) = \frac{(x-4)^2}{x^2}$
 - (a) Find $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow 0^-} f(x)$. Deduce the vertical asymptote of $f(x)$.
(4 marks)
 - (b) Find $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$. Deduce the horizontal asymptote of $f(x)$.
(4 marks)
2.
 - (a) Find $\frac{dy}{dx}$ for $y = x^2 \cos x^2$.
(4 marks)
 - (b) Given $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$, then show that $\frac{dy}{dx} = \frac{4}{(e^x + e^{-x})^2}$.
(4 marks)
3.
 - (a) Find the critical number of $f(x) = \sqrt{x^2 - 16}$.
(5 marks)
 - (b) Find the interval(s) on which f is increasing and decreasing.
(4 marks)





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SECTION B [75 marks]
BAHAGIAN B [75 markah]

This section consists of 7 questions. Answer all questions.
Bahagian ini mengandungi 7 soalan. Jawab semua soalan.

1. A complex number, z , is such that $|z-1| = 2\sqrt{2}$ and $\arg(z-1) = -\frac{1}{4}\pi$. Express z in the form $a+bi$, where a and b are real constants. (5 marks)

- 2 (a) Find the only real solution of the logarithmic equation
 $2\log_2 x + \log_2(x-1) - \log_2(5x+4) = 1$. (5 marks)

- (b) (i) Given $|t| = 3$, find the possible values of $|2t-1|$. (3 marks)

- (ii) Solve the inequality $|x-\sqrt{2}| > |x-3\sqrt{2}|$. (3 marks)

3. Given that f, g, h are as follows :

$$f(x) = |x|$$

$$g(x) = x^2 - 2$$

$$h(x) = \frac{1}{x}, \quad x \neq 0$$

- i) Find $F(x) = f \circ g \circ h(x)$ (3 marks)

- ii) Find the values of x when $F(x) = 5$ (3 marks)

- 4 (a) Given that $f : x \mapsto \ln(3x-2) + 5$,

- i) Find f^{-1} , and state its domain and range. (4 marks)

- ii) Hence, show the relationship between the graphs of f and f^{-1} on a diagram.

(3 marks)





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- (b) The function g and h are defined by $g(x) = \frac{x+1}{x-2}$ and $h(x) = \frac{ax+3}{x}$. Given that $hg^{-1}(4) = 6$, calculate the value of a .

(7 marks)

5. (a) Find the vertical and horizontal asymptotes of the graph of $f(x) = \frac{4x}{x^3+8}$.

(6 marks)

- (b) The function f is defined by

$$f(x) = \begin{cases} \frac{x^2-4}{x-2} & , \quad x < 2 \\ a^2-2 & , \quad x = 2 \\ \frac{|2-x|}{x-2} + b & , \quad x > 2 \end{cases}$$

where a and b are constants.

- i) If $\lim_{x \rightarrow 2} f(x)$ exists, find the value of b . (5 marks)
- ii) If $f(x)$ is continuous at $x = 2$, determine the value of a . (3 marks)

- 6) An inverted conical tank with radius 3 meter and height 4 meter is fully filled with grain. The grain is leaking from the vertex of the tank at a rate of $0.2 \text{ m}^3 \text{ s}^{-1}$. At the instant when the height of grain is 2 meters, calculate the rate of change of:

- (a) the water level. (7 marks)
- (b) the radius of the water surface. (2 marks)





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7. (a) Given $2y^2 - 3xy + x^2 = 2$.

i) By using implicit differentiation, show that:

$$(-3y + 2x) \frac{d^2x}{dy^2} + 2 \left(\frac{dx}{dy} \right)^2 - 6 \frac{dx}{dy} + 4 = 0 \quad (4 \text{ marks})$$

ii) Find the values of $\frac{d^2x}{dy^2}$ when $x = 0$. (5 marks)

(b) Given the parametric equations $x = 2 \cos^3 t$ and $y = 4 \sin^3 t$.

i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$. (4 marks)

ii) Find the exact value of $\frac{d^2y}{dx^2}$ when $t = \frac{\pi}{3}$. (3 marks)

END OF QUESTION PAPER





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ANSWERS SCHEME

SECTION A [25 marks]
BAHAGIAN A[25 markah]

NO	ANSWER SCHEME	MARKS
1	a) $f(x) = \frac{(x-4)^2}{x^2}$ $\lim_{x \rightarrow 0^+} \frac{(x-4)^2}{x^2} = +\infty$ $\lim_{x \rightarrow 0^-} \frac{(x-4)^2}{x^2} = +\infty$ $\therefore$ vertical asymptote of f is at $x = 0$ b) $\lim_{x \rightarrow +\infty} \frac{x^2 - 8x + 16}{x^2}$ $\lim_{x \rightarrow -\infty} \frac{x^2 - 8x + 16}{x^2}$ $\lim_{x \rightarrow +\infty} \frac{1 - \frac{8}{x} + \frac{16}{x^2}}{1} = 1$ $\lim_{x \rightarrow -\infty} \frac{1 - \frac{8}{x} + \frac{16}{x^2}}{1} = 1$ $\therefore$ Horizontal asymptote of f is $y = 1$	K1 J1J1 B1 K1 J1J1 B1
	Total	8 marks
2	a) $y = x^2 \cos x^2$. Let $u = x^2$, $v = \cos x^2$ $u' = 2x$, $v' = -2x \sin x^2$ $\frac{dy}{dx} = -2x^3 \sin x^2 + 2x \cos x^2$ $= 2x(\cos x^2 - x^2 \sin x^2)$ b) $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$	K1 J1 K1 J1





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	$\frac{dy}{dx} = \frac{(e^x + e^{-x}) \frac{d}{dx}(e^x - e^{-x}) - (e^x - e^{-x}) \frac{d}{dx}(e^x + e^{-x})}{(e^x + e^{-x})^2}$ $= \frac{(e^x + e^{-x})(e^x + e^{-x}) - (e^x - e^{-x})(e^x - e^{-x})}{(e^x + e^{-x})^2}$ $= \frac{(e^x + e^{-x})^2 - (e^x - e^{-x})^2}{(e^x + e^{-x})^2}$ $= \frac{(e^x + e^{-x} - e^x + e^{-x})(e^x + e^{-x} + e^x - e^{-x})}{(e^x + e^{-x})^2}$ $= \frac{4}{(e^x + e^{-x})^2}$	K1 K1 K1 J1												
	Total	8 Marks												
3 a)	Domain: $(-\infty, -4] \cup [4, \infty)$ $f'(x) = \frac{1}{2}(x^2 - 16)^{-\frac{1}{2}}(2x)$ $= \frac{x}{\sqrt{x^2 - 16}}$ $f'(x) = 0$ $x = 0$ <i>not in the domain</i> $f'(x)$ undefined $x^2 - 16 = 0$ $(x + 4)(x - 4) = 0$ $x = -4, x = 4$	J1 K1 J1 K1 J1												
b)	<table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <tr> <td>Interval</td><td>$(-\infty, -4)$</td><td>$(4, \infty)$</td></tr> <tr> <td>Test value</td><td>-5</td><td>5</td></tr> <tr> <td>Sign of f'</td><td>-ve</td><td>+ve</td></tr> <tr> <td></td><td>\</td><td>/</td></tr> </table> $(-\infty, -4)$ decreasing $(4, \infty)$ increasing	Interval	$(-\infty, -4)$	$(4, \infty)$	Test value	-5	5	Sign of f'	-ve	+ve		\	/	K1 K1 J1 J1
Interval	$(-\infty, -4)$	$(4, \infty)$												
Test value	-5	5												
Sign of f'	-ve	+ve												
	\	/												





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	Total	9 marks
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ANSWERS SCHEME

SECTION B [75 marks]

BAHAGIAN B [75 markah]

NO	ANSWER SCHEME	MARKS
1	Method 1 $ z - 1 = 2\sqrt{2}$ $\arg(z - 1) = -\frac{1}{4}\pi$ $z - 1 = 2\sqrt{2} \left[\cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right) \right]$ $= 2\sqrt{2} \left[\frac{\sqrt{2}}{2} + i \left(-\frac{\sqrt{2}}{2}\right) \right]$ $= 2 - 2i$ $\therefore z = 3 - 2i$ Method 2 Let $z = a + bi$ $ z - 1 = 2\sqrt{2}$ $ (a + bi) - 1 = 2\sqrt{2}$ $ (a - 1) + bi = 2\sqrt{2}$ $\sqrt{(a - 1)^2 + b^2} = 2\sqrt{2}$ $(a - 1)^2 + b^2 = 8 \text{----- (1)}$	B1 K1✓ K1✓ J1 J1 B1 K1✓ K1✓





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	$\arg(a + bi - 1) = -\frac{\pi}{4}$ $\arg(a - 1 + bi) = -\frac{\pi}{4}$ $\tan^{-1}\left(\frac{b}{a-1}\right) = -\frac{\pi}{4}$ $\frac{b}{a-1} = \tan\left(-\frac{\pi}{4}\right)$ $\frac{b}{a-1} = -1$ $b = 1 - a \text{ --- (2)}$ Substitute (2) into (1) $b^2 + b^2 = 8$ $b^2 = 4$ $b = \pm 2$ $\therefore b = -2 \text{ since } \arg(z-1) < 0, \therefore a = 1 - (-2) = 3$ Hence $z = 3 - 2i$	J1 J1
	Total	5 Marks
2 (a)	$2\log_2 x + \log_2(x-1) - \log_2(5x+4) = 1.$ $\log_2 \frac{x^2(x-1)}{(5x+4)} = 1$ $\frac{x^2(x-1)}{(5x+4)} = 2$ $x^3 - x^2 - 10x - 8 = 0$ $(x+1)(x+2)(x-4) = 0$ $x = -1(\text{reject}), x = -2(\text{reject}), x = 4$ $x = 4.$	K1 K1 K1 J1 J1
(b) i)	$ t = 3 \Rightarrow t = -3, t = 3$ $\text{When } t = -3, 2(-3) - 1 = -7 = 7$ $\text{When } t = 3, 2(3) - 1 = 5 = 5$	K1 K1J1





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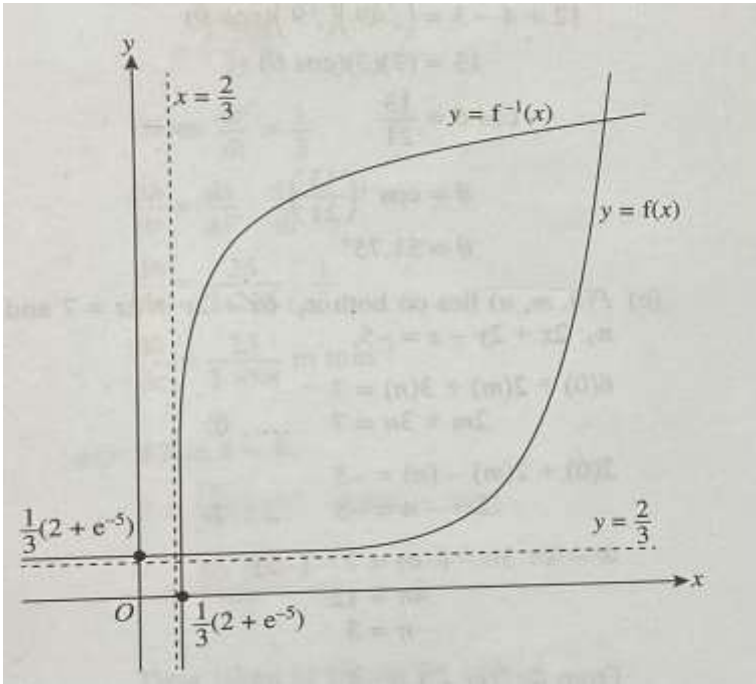
ii)	$ x - \sqrt{2} > x + 3\sqrt{2} $ $\left(x - \sqrt{2} \right)^2 > \left(x + 3\sqrt{2} \right)^2$ $(x - \sqrt{2})^2 > (x + 3\sqrt{2})^2$ $x^2 - 2\sqrt{2}x + 2 > x^2 + 6\sqrt{2}x + 18$ $x < -\sqrt{2}$ $\{x : x < -\sqrt{2}\}$	K1 K1 J1
	Total	11 Marks
3	(i) $F(x) = f \circ g \circ h(x)$ $= f \circ g\left(\frac{1}{x}\right)$ $= f\left(\frac{1}{x^2} - 2\right)$ $= \left \frac{1}{x^2} - 2\right $ $= \left \frac{1 - 2x^2}{x^2}\right $ (ii) $\left \frac{1 - 2x^2}{x^2}\right = 5$ $\frac{1 - 2x^2}{x^2} = 5 \qquad \frac{1 - 2x^2}{x^2} = -5$ $1 - 2x^2 = 5x^2 \qquad 1 - 2x^2 = -5x^2$ $-7x^2 = -1 \qquad 3x^2 = -1$ $x = \pm\sqrt{\frac{1}{7}} \qquad x^2 = -\frac{1}{3}$ no real root $\therefore x = x = \pm\sqrt{\frac{1}{7}}$	K1 K1 J1 B1 K1√ J1
	Total	6 Marks
	(i)	





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4(a)	(i) $f(x) = \ln(3x - 2) + 5, \quad x > \frac{2}{3}$ $f[f^{-1}(x)] = x$ $\ln(3f^{-1}(x) - 2) + 5 = x$ $e^{\ln(3f^{-1}(x) - 2)} = e^{x-5}$ $3f^{-1}(x) - 2 = e^{x-5}$ $f^{-1}(x) = \frac{e^{x-5} + 2}{3}$ $D_{f^{-1}} : x \in \mathbb{R}$ $R_{f^{-1}} : y > \frac{2}{3}, y \in \mathbb{R}$ (ii)  (b) $gg^{-1}(x) = x$ $\frac{g^{-1}(x) + 1}{g^{-1}(x) - 2} = x$ $g^{-1}(x) + 1 = x[g^{-1}(x) - 2]$ $g^{-1}(x) = \frac{2x + 1}{x - 1}$	B1 K1 J1 J1 (both) R1 R1 R1 B1 K1 J1
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	$g^{-1}(4) = \frac{2(4)+1}{4-1} = 3$ $h(3) = 6$ $\frac{3a+3}{3} = 6$ $a = \frac{18-3}{3}$ $a = 5$	K1J1 K1 J1
	Total	14 Marks
5 a)	$f(x) = \frac{4x}{x^3+8}$ Since $x^3+8 \neq 0$, then $D_f : \mathbb{R} \setminus \{-2\}$. The vertical asymptote, $x = -2$. Examine the one-sided limits $\lim_{x \rightarrow -2^+} \frac{4x}{x^3+8} = -\infty \quad , \quad \lim_{x \rightarrow -2^-} \frac{4x}{x^3+8} = +\infty$ Then $x = -2$ is the vertical asymptote. Horizontal asymptotes, $y = \lim_{x \rightarrow +\infty} \frac{4x}{x^3+8}$ $= \lim_{x \rightarrow +\infty} \frac{\frac{4}{x^2}}{1 + \frac{8}{x^3}}$ $= \frac{0}{1+0}$ $= 0$ $y = \lim_{x \rightarrow -\infty} \frac{4x}{x^3+8}$ $= \lim_{x \rightarrow -\infty} \frac{\frac{4}{x^2}}{1 - \frac{8}{x^3}}$ $= \frac{0}{1-0}$ $= 0$ Then $y = 0$ is the horizontal asymptote.	K1 K1 J1 K1 J1 J1
5 b)		
(i)		K1





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(ii)	$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$ $\lim_{x \rightarrow 2^-} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2^+} \frac{ 2 - x }{x - 2} + b$ $\lim_{x \rightarrow 2^-} \frac{(x - 2)(x + 2)}{x - 2} = \lim_{x \rightarrow 2^+} \frac{-(2 - x)}{x - 2} + b$ $\lim_{x \rightarrow 2^-} (x + 2) = \lim_{x \rightarrow 2^+} 1 + b$ $4 = 1 + b$ $b = 3$ $f(2) = a^2 - 2$ Since $f(x)$ is continuous at $x = 2$, then $\lim_{x \rightarrow 2} f(x) = f(2)$ $4 = a^2 - 2$ $a^2 = 6$ $a = \pm\sqrt{6}.$	K1K1 K1 J1 K1 K1 J1
	Total	14 marks
6 (a)	$\frac{r}{3} = \frac{h}{4}$ $r = \frac{3}{4}h$ $V = \frac{1}{3}\pi r^2 h$ $= \frac{1}{3}\pi \left(\frac{3}{4}h\right)^2 h$ $= \frac{3}{16}\pi h^3$ $\frac{dV}{dh} = \frac{9}{16}\pi h^2$ $\frac{dh}{dV} = \frac{16}{9\pi h^2}$ $\frac{dh}{dt} = \frac{dh}{dV} \cdot \frac{dV}{dt}$ $= \frac{16}{9\pi h^2} \cdot (-0.2)$ $= -\frac{16}{45\pi h^2}$	J1 K1 K1 K1 J1





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	$\text{at } h = 2,$ $\frac{dh}{dt} = -\frac{16}{45\pi(2)^2}$ $= -\frac{4}{45\pi}$	K1 J1
6 (b)	$r = \frac{3}{4}h$ $\frac{dr}{dt} = \frac{3}{4} \frac{dh}{dt}$ $\text{at } h = 2,$ $\frac{dh}{dt} = -\frac{4}{45\pi}$ $\frac{dr}{dt} = \frac{3}{4} \cdot -\frac{4}{45\pi}$ $= -\frac{1}{15\pi}$	K1 J1
	Total	9 marks
7 (a) i	$2y^2 - 3xy + x^2 = 2$ $4y - 3\left[x \cdot 1 + y \frac{dx}{dy}\right] + 2x\left(\frac{dx}{dy}\right) = 0$ $4y - 3x - 3y \frac{dx}{dy} + 2x \frac{dx}{dy} = 0$ $4 - 3 \frac{dx}{dy} - 3\left[y \cdot \frac{d^2x}{dy^2} + \frac{dx}{dy} \cdot 1\right] + 2\left[x \cdot \frac{d^2x}{dy^2} + \left(\frac{dx}{dy}\right)^2\right] = 0$ $4 - 3 \frac{dx}{dy} - 3y \frac{d^2x}{dy^2} - 3 \frac{dx}{dy} + 2x \frac{d^2x}{dy^2} + 2\left(\frac{dx}{dy}\right)^2 = 0$ $(-3y + 2x) \frac{d^2x}{dy^2} + 2\left(\frac{dx}{dy}\right)^2 - 6 \frac{dx}{dy} + 4 = 0 \quad (\text{shown})$	K1 K1 K1 J1
(a) ii	$\text{when } x = 0,$ $2y^2 - 0 + 0 = 2$ $y^2 = 1$ $y = \pm 1$	K1 K1





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	when $y = 1$, $\frac{dx}{dy} = \frac{4}{3}$ when $y = -1$, $\frac{dx}{dy} = \frac{4}{3}$ when $x = 0, y = 1, \frac{dx}{dy} = \frac{4}{3}$, $(-3)\frac{d^2x}{dy^2} + 2\left(\frac{4}{3}\right)^2 - 6\left(\frac{4}{3}\right) + 4 = 0$ $\frac{d^2x}{dy^2} = -\frac{4}{27}$ when $x = 0, y = -1, \frac{dx}{dy} = \frac{4}{3}$, $(3)\frac{d^2x}{dy^2} + 2\left(\frac{4}{3}\right)^2 - 6\left(\frac{4}{3}\right) + 4 = 0$ $\frac{d^2x}{dy^2} = \frac{4}{27}$	J1 J1 J1
7 (b)i	$x = 2\cos^3 t \qquad y = 4\sin^3 t$ $\frac{dx}{dt} = 6\cos^2 t \cdot (-\sin t) \qquad \frac{dy}{dt} = 12\sin^2 t \cdot (\cos t)$ $= -6\cos^2 t \sin t \qquad = 12\sin^2 t \cos t$ $\frac{dy}{dx} = \frac{12\sin^2 t \cos t}{-6\cos^2 t \sin t}$ $= -2 \tan t$	K1 J1





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$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$ $= \frac{\frac{d}{dt}(-2 \tan t)}{-6 \cos^2 t \sin t}$ $= \frac{-2 \sec^2 t}{-6 \cos^2 t \sin t}$ $= \frac{1}{3 \cos^4 t \sin t}$ when $t = \frac{\pi}{3}$ $\frac{d^2y}{dx^2} = \frac{1}{3 \left[\cos\left(\frac{\pi}{3}\right) \right]^4 \sin\left(\frac{\pi}{3}\right)}$ $= \frac{1}{3 \left[\frac{1}{2} \right]^4 \frac{\sqrt{3}}{2}}$ $= \frac{1}{3 \left(\frac{1}{16} \right) \frac{\sqrt{3}}{2}}$ $= \frac{1}{\left(\frac{3}{32} \right) \sqrt{3}}$ $= \frac{32}{3\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}}$ $= \frac{32}{9} \sqrt{3}$	K1 J1 K1 K1 (conjugate) J1
Total	16 Marks

